

MAX HEAP SORT:

Max Heap Sort is a comparison-based sorting algorithm that first creates a Max Heap, where the largest element is at the root. The largest element is then repeatedly moved to the end of the array

ALGORITHM:

1. Start with the given array.
2. Build a Max Heap from the array.
3. The largest element will be at the root ($\text{arr}[0]$).
4. Swap the root with the last element.
5. Reduce the heap size by one.
6. Apply Heapify to restore the Max Heap.
7. Repeat until all elements are sorted.

Time Complexity:

Case	Complexity
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Best	$O(n \log n)$
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Average	$O(n \log n)$
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Worst	$O(n \log n)$
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Space Complexity:

Space Complexity = **$O(1)$**

OUTPUT:

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Enter the number of elements: 4
Enter the element 1: 154
Enter the element 2: 2
Enter the element 3: 100
Enter the element 4: 48
Sorted array: 2 48 100 154

...Program finished with exit code 0
Press ENTER to exit console.□
```

CONCLUSION:

Max Heap Sort provides a guaranteed $O(n \log n)$ running time and does not require extra array space. It is useful when consistent performance and low auxiliary memory usage are important.